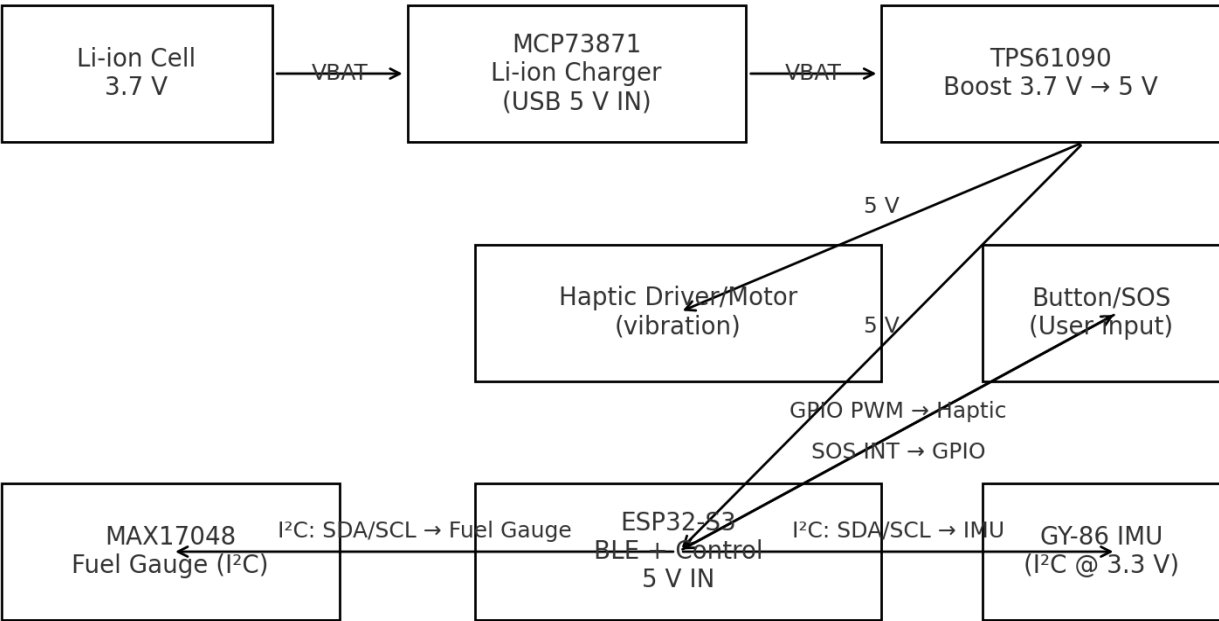


Smart Belt – Block Diagram & System Schematic

High-level block diagram and conceptual textbook-style wiring schematic with reference notes and suggested component values.

Block Diagram

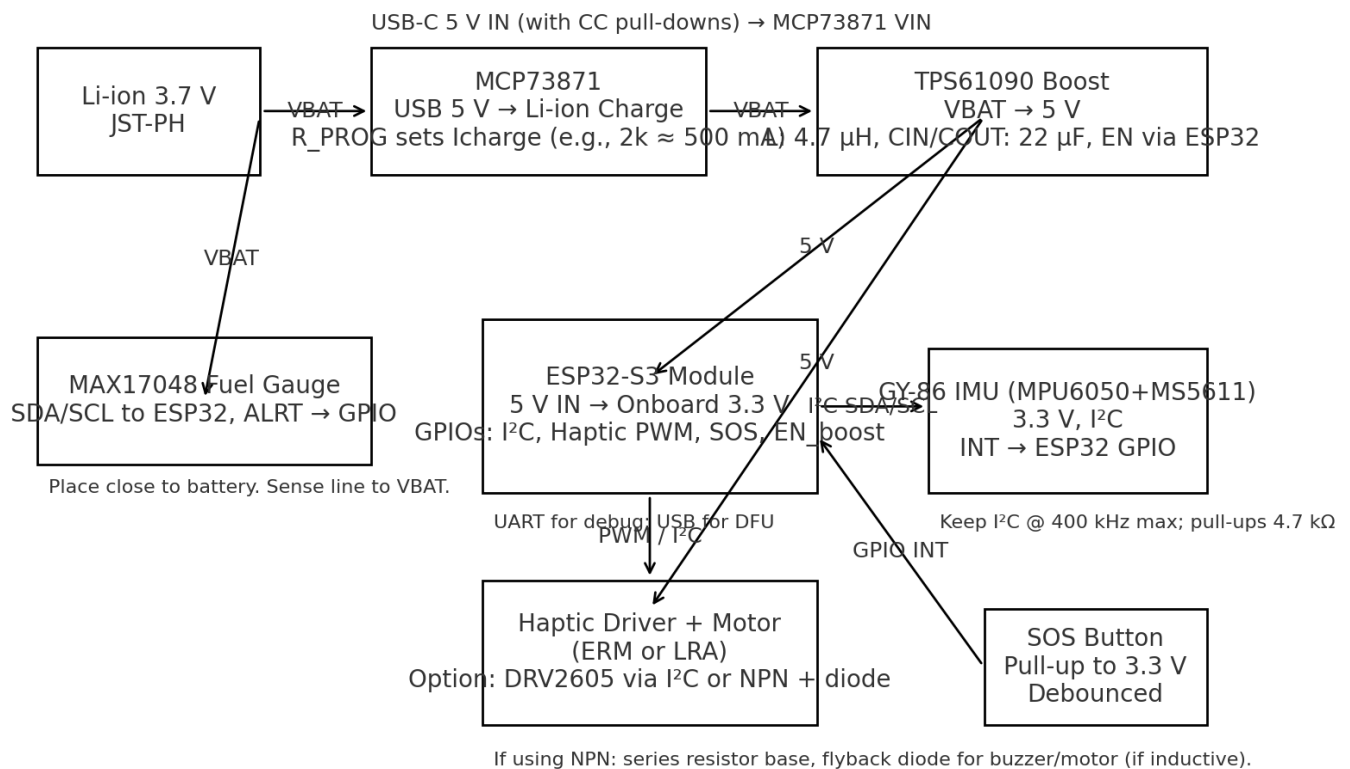
SMART BELT – BLOCK DIAGRAM



Notes: Use 3.3 V LDO for ESP32-S3 & sensors internally; 5 V bus from booster feeds modules requiring 5 V. Level-shift I²C if needed.

System Schematic (Conceptual Wiring)

SMART BELT - SYSTEM SCHEMATIC (Conceptual Wiring)



Common GND

Bill of Materials (Prototype)

#	Component	Part / Notes
1	MCU / BLE	ESP32-S3 module (e.g., WaveShare ESP32-S3)
2	IMU	GY-86 (MPU6050 + MS5611). I _{2C} @ 3.3 V
3	Battery	Li-ion 3.7 V (JST-PH) with protection
4	Charger	MCP73871. Set R_PROG for desired I _{charge}
5	Booster	TPS61090 3.7 V → 5 V. L ≈ 4.7 μH, C _{IN} /C _{OUT} ≈ 22 μF
6	Fuel Gauge	MAX17048 (I _{2C}) + ALRT pin to ESP32
7	Haptic	DRV2605 (I _{2C}) + ERM/LRA motor OR NPN + diode
8	Button	SOS momentary switch to GPIO with pull-up
9	Passives	I _{2C} pull-ups ~4.7 kΩ; decoupling 0.1 μF near ICs